

See Through the Inside and Outside: Human Body and Anatomical Skeleton Prediction Network

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ABSTRACT

In this work, it is the first time to use a monocular image with an arbitrary human pose to predict the human body and anatomical skeleton simultaneously. The framework, called STIO (See Through the Inside and Outside), considering the human anatomical constraints, consists of a two-way reconstruction network that forms a strong collaboration between skeleton and skin.

Index Terms: Computing methodologies—Artificial intelligence—Computer vision—Computer vision tasks; Computing methodologies—Artificial intelligence—Computer vision—Computer vision problems

1 INTRODUCTION AND RELATED WORK

How to reconstruct the human body from an RGB image and infer complex skeletal kinematics changes to achieve skeletal modeling with different infusions is an in-depth topic spanning kinematics, robotics, anatomy, and medicine. On many non-medical occasions, there is no motion capture equipment or X-ray machine, such as AR games, massage therapy, physiological teaching, *etc.* But it is expected that the body and skeleton information can be predicted in a low-cost way. The human body reconstruction methods [2, 4–6, 9, 10] based on regression can generate human parameters directly from the extracted features but need a lot of data supervision. In contrast, the optimization-based methods [1] are accurate but sensitive to the initial value and require large time consumption. The result of most skeleton prediction methods, which is commonly represented as a simplified bone model composed of 2D or 3D joint points, is inconsistent with the real skeleton, thus hindering the application in the above mentioned fields. The state-of-art anatomical skeleton prediction method [3] requires a human body 3D model in advance, and may introduce cumulative errors. In this work, we present STIO, a two-way regression framework with optimized supervision, which allows the reconstruction of the anatomical skeleton and human mesh with the corresponding pose and shape. To the best of our knowledge, we are the first to infer the human body and anatomical skeleton simultaneously of a person with an arbitrary pose from the monocular image, *i.e.* we predict both the inside (bones) and the outside (skin). For the prediction of the anatomical skeleton, we propose a new single-stage optimization to speed up self-supervision and correlate two-way optimization pathways with each other to avoid statistical correlation learning failures. Experimental results show that the proposed method outperforms the relevant method in accuracy, time consumption, and convergence speed.

2 METHODOLOGY

Our framework is as shown in the Fig 1: Input a monocular RGB image, the common features of the human body and skeleton are ex-

tracted through the shared *Feature Extraction* layer. Then a two-way branch is formed through the HTS (human body to skeleton) module, as shown in the light green area. HTS uses the two-way branches to predict the parameters required by the human body and anatomical skeleton model respectively. This part is called the *Regression* process. After obtaining the initial parameters, we optimize these parameters respectively through the *Optimization* process, as shown in the orange area. The optimized result is used as the training supervision (purple area) of the regression, *i.e.* the *Regression Supervision* part. In this part, the predicted human body and skeleton parameters are matched to the input image. The two branches supervise each other and make the regression result improve cooperatively. With the iteration of training, the results of the regression become more accurate, and the optimization can get better initial values to supervise the regression process, forming an effective positive iteration. In this work, SMPL [7] and stitched puppet [8] with anatomical skeleton models are used to represent human and skeleton 3D models.

2.1 Regression

As shown in the HTS module, the initial model parameters are obtained through multiple regression iteration. Let θ represent human body parameters including pose p , camera parameter c and shape β_{body} , ϵ represent the skeleton parameter which consists of translation t and rotation r . The loss function of the regression part can be expressed as:

$$L_{hum} = \lambda_{hum} \|J_{reg} - J_{opt}\| + \lambda_{\theta} \|\theta_{reg} - \theta_{opt}\| \quad (1)$$

$$L_{skel} = \lambda_{ske} \|\mathcal{L}_{skel} - \mathcal{L}_{body}\| + \lambda_{\epsilon} \|\epsilon_{reg} - \epsilon_{opt}\| \quad (2)$$

where L_{hum} is the human body branch loss, J_{reg} represents the 2D joints position obtained by reprojecting the 3D joints of human body to the image using the predicted camera parameters, J_{opt} is provided by the following optimization step and can be regarded as the ground truth of this round of training. θ_{reg} and θ_{opt} are human body parameters before and after optimization. L_{skel} is the skeleton branch loss, $\mathcal{L}_{skel}$ represents the joints obtained from the skeleton and $\mathcal{L}_{body}$ means the 3D joints inferred from the human body mesh. ϵ_{reg} and ϵ_{opt} are similar with θ term in L_{hum} .

2.2 Optimization

In the optimization part, we use SMPLify [1] to optimize the human body parameters, so that the obtained 3D human body model is more consistent with the original image. After optimization, human body parameters and joint position can be used as the supervision of the regression process, which has been mentioned above. For skeleton optimization, we propose OSF⁺ algorithm on the basis of OSSO [3], which greatly optimizes its time consumption for network training. The objective function can be written as:

$$\begin{aligned} E(t, r) = & \lambda_l \|\mathcal{L}_{skel}(SK(\beta_{skel}, t, r)) - \mathcal{L}_{body}(M(\beta_{body}, p))\| \\ & + \lambda_{ct} E_{ct}(\beta_{skel}, t, r; SK_0) + \lambda_j E_j(\beta_{skel}, t, r; SK_0) \\ & + \lambda_{clv} E_{clv}(\beta_{skel}, t, r; SK_0) \end{aligned} \quad (3)$$

where $M(\beta_{body}, p)$ means human body model, $SK(\beta_{skel}, t, r)$ is the anatomical skeleton model, $SK_0 = SK(\beta_{skel}, t_0, r_0)$ represents the

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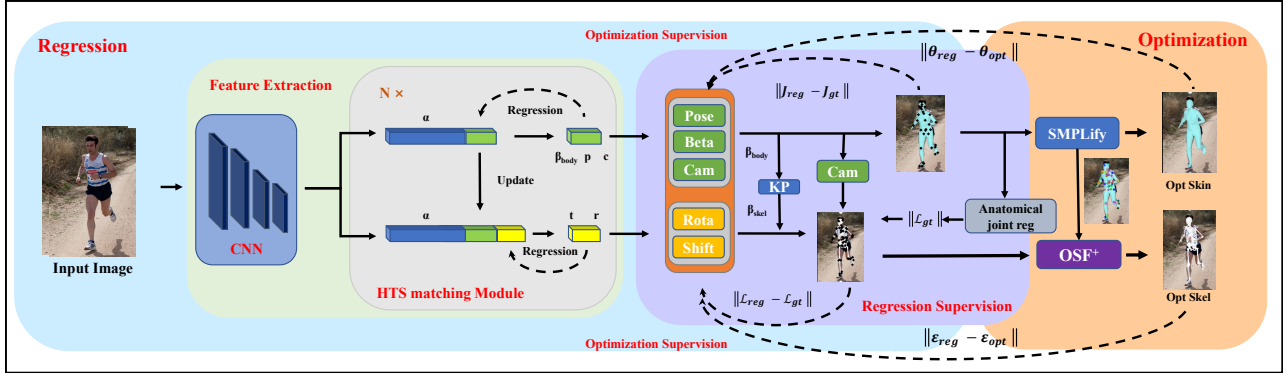


Figure 1: Overview of our proposed method.

skeleton with the initial standard T -pose and the inferred shape $\beta_{skel} = KP(\beta_{body})$. $KP(\cdot)$ is the shape transition regressor which is trained before. $E_{cl}(\cdot)$ is the anatomical constraint which is designed of more than 3,000 points to restrain the relative distance of skeletons [3]. E_j is the punishment term to make the connection distance between different bones keep the same. E_{cly} means the anatomical constraints of the clavicle relative to the thorax. λ means the weight coefficients.

3 EXPERIMENTS

We conduct quantitative experiments on the human body and skeleton results, as shown in the Figure 2. Compared with the original skeleton algorithm [3], the proposed STIO method can achieve a speed increase of more than 700 times ($140.8 / 0.2$) with the cost of minimum accuracy loss. It is worth noting that our model is only trained on a small batch of data sets, and does not use the 3DPW training set. Compared with the latest models trained on the 3DPW training set, our method still obtains exciting results in the field of 3D human reconstruction. The visualization results are shown in the Figure 3.

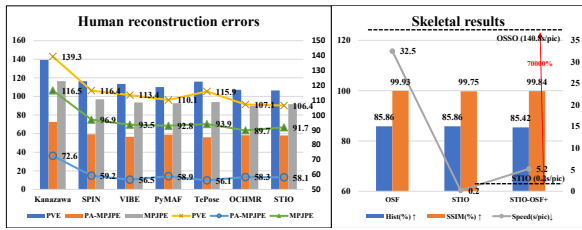


Figure 2: Performed result of the human body and skeleton.

4 DISCUSSION AND CONCLUSION

In this work, a two-way regression framework with optimized supervision is proposed to predict the human body and anatomical skeleton simultaneously. The network considers the human anatomical constraints and forms a tight connection between the human body and the anatomical skeleton. However, when meet some more rigorous fields such as medicine treatment, effective supervision of large quantities of real medical data is needed. In the future, we will combine more real medical data to improve this limitation.

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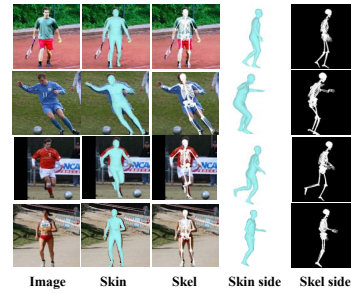


Figure 3: Examples of STIO result. Other views of the human body and skeleton are also given.

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